higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T1440**(E)**(A15)T

NATIONAL CERTIFICATE

QUANTITY SURVEYING N4

(2050004)

15 April 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: **Dimension paper (OE 8/12)**
Abstract paper (OE 8/10)
Billing paper (OE 8/11)

Candidates are NOT allowed to use the STANDARD SYSTEM OF MEASURING BUILDING WORK.

Calculators may be used.

This question paper consists of 6 pages and 1 addendum.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
QUANTITY SURVEYING N4
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. SECTION A must be done on the prescribed sheets provided.
5. SECTION B must be done in the ANSWER BOOK.
6. Sketches must be large, neat and fully labelled.
7. Do NOT use RED or GREEN ink.
8. The use of tippex is not allowed.
9. Write neatly and legibly.

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SECTION A

QUESTION 1

The ADDENDUM (attached) shows a floor plan and a cross-sectional view through the foundation of a small dwelling.

Measure the foundation up to and including the concrete surface bed as instructed below.

SPECIFICATIONS

Earthwork: - Excavations in firm ground
- Excavated earth to be used for filling
- Site clearance 2 metres from external walls

Concrete: - 15 MPa

Masonry: - Plaster bricks in 1:3 cement mortar
- Face bricks to two courses below ground level
- One course 75 mm

1.1	1.1.1	Calculate the length and width for the site clearance.	(1)
	1.1.2	Take off the site clearance item.	(3)
1.2	1.2.1	Calculate the centre line of the 650 mm wide footing.	(1)
	1.2.2	Calculate the excavation depth.	(1)
	1.2.3	Take off the item for the foundation excavations.	(3)
	1.2.4	Take off the item for risk of collapse.	(2)
	1.2.5	Take off the item for backfilling.	(3)
1.3	1.3.1	Take off the item for the concrete foundation footing.	(3)
	1.3.2	Take off the items for the concrete surface bed.	(6)
1.4	1.4.1	Take off the item for the 330 mm thick foundation wall.	(3)
	1.4.2	Take off the item for the facings.	(4)

[30]

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QUESTION 2

Prepare a tender document for the earthwork trade as follows:

2.1 Square (2½)

2.2 Abstracting (3½)

2.3 Billing (4)

[10]

TOTAL SECTION A: 40

SECTION B

QUESTION 3

The quantity surveyor needs to make provision in the 'take off' for a steel door frame once rebated, suitable for a 110 mm wall and a 913 × 2 032 × 44 mm flush timber door. Adjustments must be made for 100 mm concrete footing, 30 mm thick cement mortar screed, 19 mm thick plaster on both sides and two coats of PVA paint on plastered walls.

Draw up a measuring list of all the items that must be measured for this opening. **[10]**

QUESTION 4

Explain the meaning of the following quantity surveying terms:

4.1 Beam filling

4.2 Labour only

4.3 Ditto

4.4 Measured net

4.5 Purpose-made

4.6 Nil

4.7 Fair facing

4.8 Striking-off and curing

4.9 Dotting-on

4.10 Casting-up

(10 × 1) **[10]**

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QUESTION 5

Write down the unit of measurement of the following items as prescribed in the Standard System of Measuring Builder's Work.

5.1 5.1.1 Keep excavations dry

5.1.2 Concrete in foundations

5.1.3 Backfilling

5.1.4 Facings

5.1.5 Foundation brickwork

(5 × 1) (5)

5.2 State FIVE items that the quantity surveyor needs to look at on site when drawing an interim valuation. (5)

[10]

QUESTION 6

Write the following abbreviations out in full:

6.1 W.p. laid in c.m. (2)

6.2 W.S.A.P tbr. (3)

6.3 Bld. in dr. frame. (2)

6.4 C.s. of all rub. and veg. matter. (3)

[10]

QUESTION 7

Write down FIVE pre-contract duties of:

7.1 The quantity surveyor

7.2 The architect

(2 × 5) **[10]**

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QUESTION 8

Make a neat sketch of a half-full bolted timber roof truss and show the following members clearly:

- Rafter
- King post
- Diagonal strut
- Tie beam
- Ridge

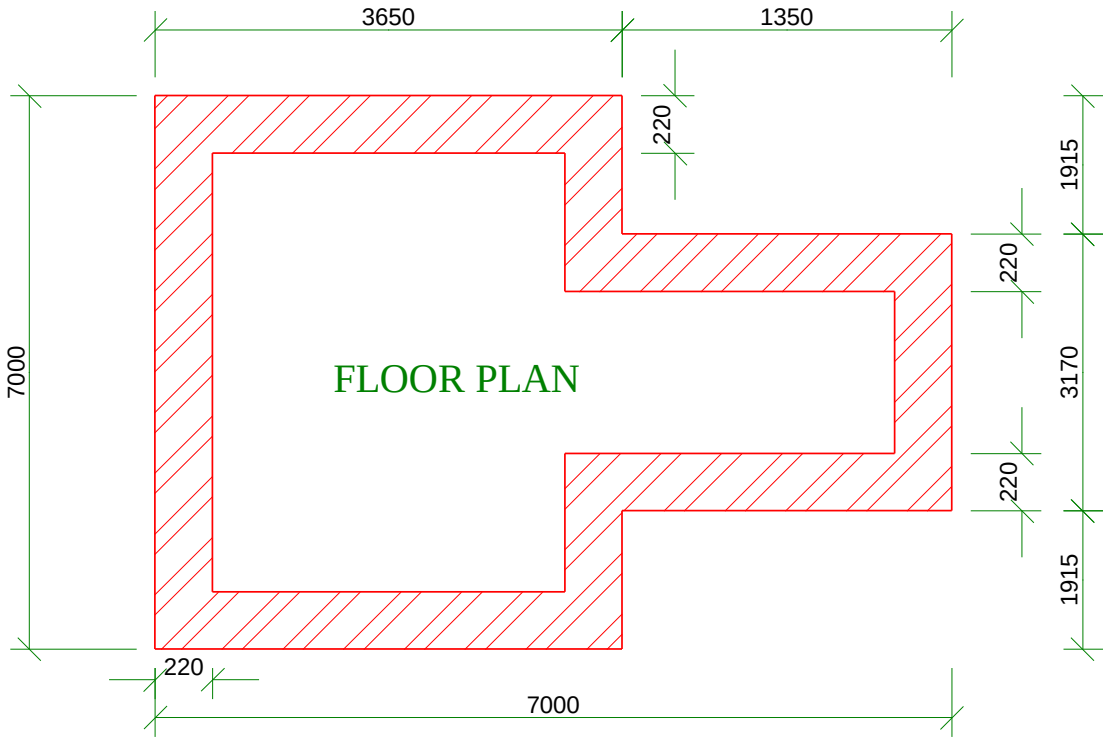
[10]

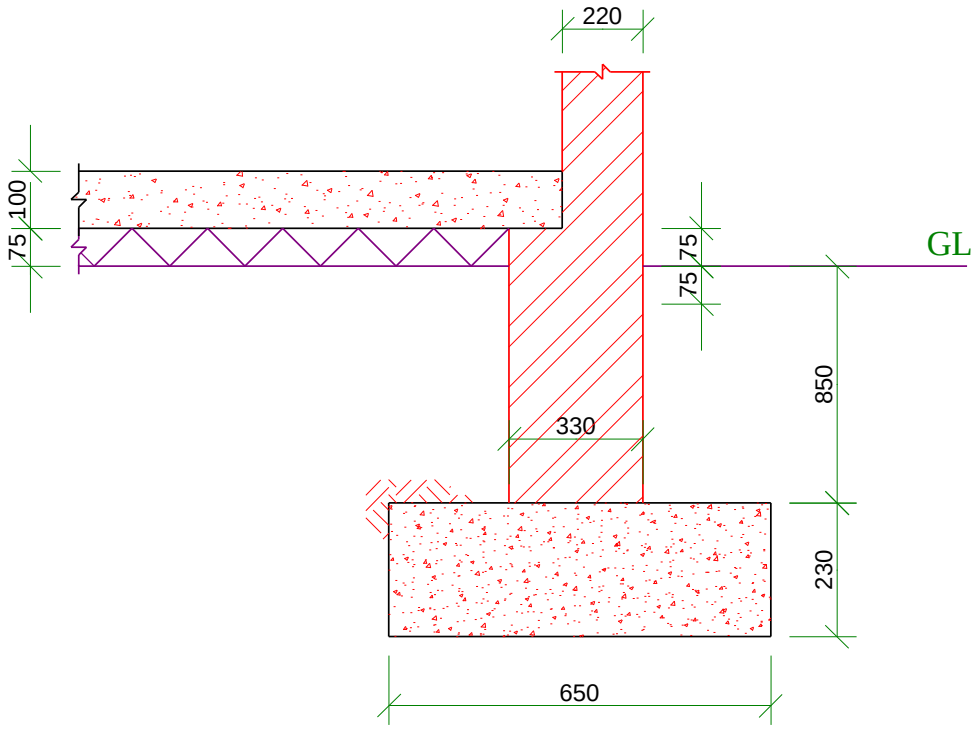
TOTAL SECTION B: 60
GRAND TOTAL: 100

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(2050004)

ADDENDUM





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